

SQL Data Analysis Project Report

Introduction

SQL (Structured Query Language) is widely used for managing and analyzing relational databases. This project demonstrates the use of SQL queries to extract, manipulate, and analyze data for decision-making purposes.

Project Objective

The main objective of this project is to perform data analysis using SQL by retrieving relevant information, applying filters, aggregating data, and generating insights.

Dataset Description

The dataset contains structured records stored in a relational database. Various SQL operations were performed to analyze the data.

Methodology

Step 1: Database Setup

- Imported the dataset into MySQL.
- Verified table structure and data types.

Step 2: Data Exploration

- Examined table contents.
- Reviewed columns and record counts.

Step 3: Query Execution

The following SQL concepts were applied:

- SELECT statements
- WHERE clause
- ORDER BY clause
- GROUP BY clause
- HAVING clause
- Aggregate Functions
- JOIN operations

Step 4: Analysis and Insights

- Identified trends and patterns.

- Calculated summary statistics.
- Generated meaningful business insights.

Results

The SQL queries successfully extracted and analyzed information from the database. The analysis provided a better understanding of the dataset and demonstrated the practical application of SQL for data analysis.

Conclusion

This project enhanced SQL skills by applying database querying techniques, aggregation functions, filtering methods, and joins to derive valuable insights from structured data.

Technologies Used

- MySQL
- SQL
- MySQL Workbench

Author

Jhansi

Data Analytics Intern – DecodeLabs